Reliability in Multiple-Burst Designs through Bursti-CTVAR

Xiaoyue Xiong[1], Charles Driver[2], Zachary Fisher[3], Sy-Miin Chow[1]

[1]The Pennsylvania State University [2]University of Zurich [3]University of North Carolina at Chapel Hill

PennState College of Health and Human Development

References

Bliese, P. D. (2000). Within-group agreement, non-independence, and reliability: Implications for data aggregation and analysis. In K. J. Klein & S. W. J. Kozlowski (Eds.), *Multilevel theory, research, and methods in organizations: Foundations, extensions, and new directions* (pp. 349–381). Jossey-Bass.

Hox, J. J., Moerbeek, M., & van de Schoot, R. (2018). *Multilevel analysis: Techniques and applications* (3rd ed.). Routledge. https://doi.org/10.4324/9781315650982

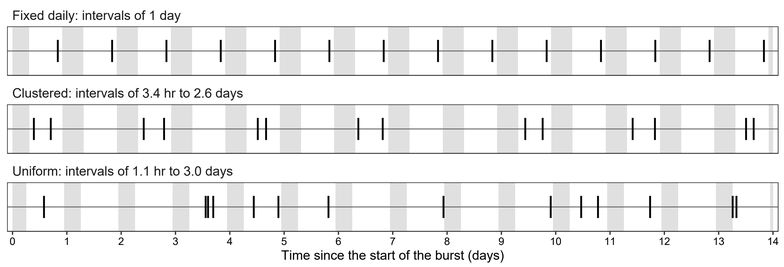
Hunter, M. D. (2026). Extending reliability to intensive longitudinal data with the Kalman filter. *British Journal of Mathematical and Statistical Psychology*. Advance online publication. https://doi.org/10.1111/bmsp.70039

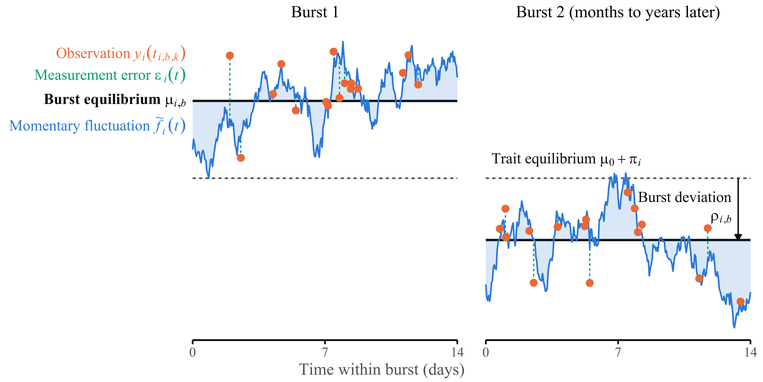
Kish, L. (1965). *Survey sampling*. Wiley.

Schuurman, N. K., & Hamaker, E. L. (2019). Measurement error and person-specific reliability in multilevel autoregressive modeling. *Psychological Methods, 24*(1), 70–91. https://doi.org/10.1037/met0000188

1. Measurement Burst Data

Occasions of one simulated person per schedule





2. The Bursti-CT-VAR Model

Measurement — equilibrium + momentary fluctuation + error:

$$\boldsymbol{y}_i(t) = \boldsymbol{L}\big(\boldsymbol{\mu}_{i,b} + \tilde{\boldsymbol{f}}_i(t)\big) + \boldsymbol{\varepsilon}_i(t), \qquad \boldsymbol{\varepsilon}_i(t) \sim N(\boldsymbol{0},\, \boldsymbol{E})$$

Within-burst dynamics — drift $\boldsymbol{A}_i$ (auto- and cross-effects), diffusion $\boldsymbol{Q}$:

$$d\tilde{\boldsymbol{f}}_i(t) = \boldsymbol{A}_i\, \tilde{\boldsymbol{f}}_i(t)\, dt + \boldsymbol{G}\, d\boldsymbol{W}_i(t), \qquad \boldsymbol{Q} = \boldsymbol{G}\boldsymbol{G}^\top$$

$$\boldsymbol{A}_i = \boldsymbol{\mu}_A + \boldsymbol{U}_i, \quad \operatorname{vec}(\boldsymbol{U}_i) \sim N\big(\boldsymbol{0}, \operatorname{diag}(\boldsymbol{\sigma}_A^2)\big), \quad \boldsymbol{A}_i \text{ stable}$$

Burst equilibria — stable trait + burst deviation:

$$\boldsymbol{\mu}_{i,b} = \boldsymbol{\mu}_0 + \boldsymbol{\beta}_b + \boldsymbol{\pi}_i + \boldsymbol{\rho}_{i,b}, \qquad \boldsymbol{\pi}_i \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_\pi), \quad \boldsymbol{\rho}_{i,b} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_{\rho_b})$$

3. Simulation Design by Research Question (RQ)

	Data	Varied
RQ1	1 variable, 1 burst; M occasions over M days	schedule × speed × N (100, 500) × M (14, 56) × model: continuous time vs. discrete time (DT-Hunter, DT-SH: occasion index as time)
RQ2	2 variables, 2 bursts; fixed daily schedule, moderate speed	persons N (100, 500) × occasions per burst M_b (8, 14, 28, 56) over M_b days
RQ3	2 bursts	N for 85% power, change of $0.2\,\sigma_y$ in the mean equilibrium, computed from the true values: 1, 2, 4 a day × 7–56 days × speed
	1 person, 1 burst	occasions for 85% power to detect inertia (carry-over), 1 a day: κ (.50–.95) × latent daily autoregression (.30, .50, .70)

Half-life (variable 1): moderate 2.0, fast 0.69 days. 100 replications; Stan.

4. Reliability Coefficients

Variance of a single observation over persons (variable v, burst b; $\boldsymbol{L} = \boldsymbol{I}$):

$$\sigma^2_{y_{vb}} = \underbrace{\sigma^2_{\pi_v}}_{\text{trait } 14\%} + \underbrace{\sigma^2_{\rho_{vb}}}_{\text{burst deviation } 21\%} + \underbrace{\big[\mathbb{E}_i \boldsymbol{P}_\infty(\boldsymbol{A}_i)\big]_{vv}}_{\text{momentary fluctuation } 52\%} + \underbrace{e_{vv}}_{\text{error } 13\%}$$

Mean momentary reliability within a person: 52% / (52% + 13%) ≈ .80

$$\bar{\kappa}_v = \mathbb{E}_i\left[\frac{[\boldsymbol{P}_\infty(\boldsymbol{A}_i)]_{vv}}{[\boldsymbol{P}_\infty(\boldsymbol{A}_i)]_{vv} + e_{vv}}\right]$$

Equilibrium reliability between persons: 14% + 21% = .35

$$\mathrm{Rel}^{\mathrm{eq}}_{v,b} = \frac{\sigma^2_{\pi_v} + \sigma^2_{\rho_{vb}}}{\sigma^2_{y_{vb}}} = \mathrm{Rel}^{\mathrm{burst}}_{v,b} + \mathrm{Rel}^{\mathrm{trait}}_{v,b}$$

Burst-deviation and trait reliability .21 and .14

$$\mathrm{Rel}^{\mathrm{burst}}_{v,b} = \frac{\sigma^2_{\rho_{vb}}}{\sigma^2_{y_{vb}}}, \qquad \mathrm{Rel}^{\mathrm{trait}}_{v,b} = \frac{\sigma^2_{\pi_v}}{\sigma^2_{y_{vb}}}$$

Reliability of the burst mean 14 daily occasions: .65 moderate, .81 fast

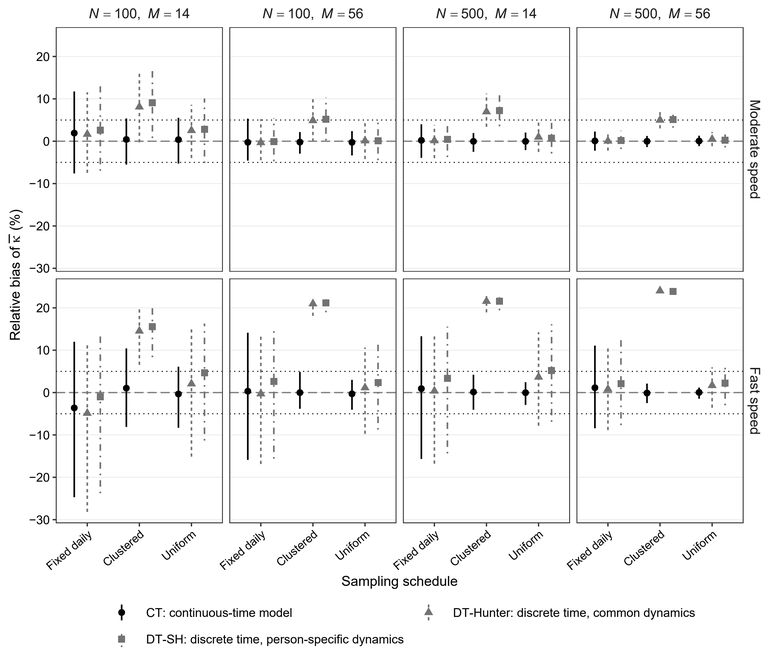
$$\mathrm{Rel}^{\bar{y}}_{v,b} = \frac{M_{\mathrm{eff}}\, \mathrm{Rel}^{\mathrm{eq}}_{v,b}}{1 + (M_{\mathrm{eff}} - 1)\, \mathrm{Rel}^{\mathrm{eq}}_{v,b}}, \qquad M_{\mathrm{eff}} = \frac{M_b}{1 + (M_b - 1)\, \vartheta_v}$$ (Hox et al., 2018; Kish, 1965)

Intraclass correlation ICC(2) 14 daily occasions: .88 moderate, .89 fast

$$\mathrm{ICC}(2)_{v,b} = \frac{M_b\, \mathrm{Rel}^{\mathrm{eq}}_{v,b}}{1 + (M_b - 1)\, \mathrm{Rel}^{\mathrm{eq}}_{v,b}}$$ (Bliese, 2000; Hox et al., 2018)

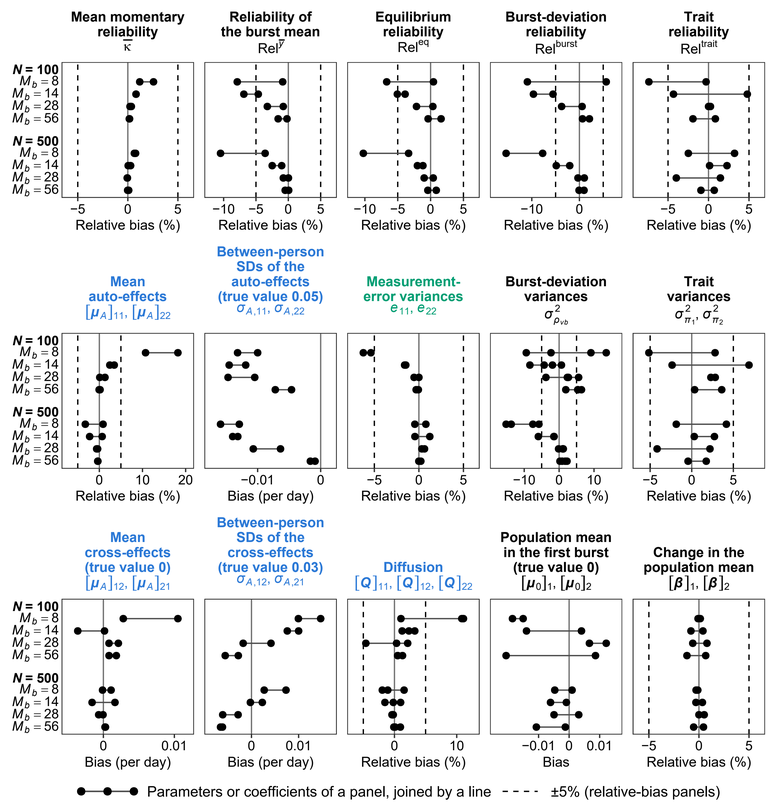
5. RQ1: What If the Occasions Are Treated as Equally Spaced?

Relative bias of mean momentary reliability, $\bar{\kappa}$



DT-Hunter (Hunter, 2026); DT-SH (Schuurman & Hamaker, 2019)

6. RQ2: Which Sample Size Configurations Suffice?



7. RQ3: What Does the Model Imply for Planning a Study?

